5G RAN

URLLC for Vehicle Communications Feature Parameter Description

Issue Draft A

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Contents

1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 05 (2025-12-14).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added support for URLLC for Vehicle Communications by UMPThb series boards. For details, see **4.3.3 Hardware**.	None	Low-frequency TDD	5900 series base stations (macro base stations) DBS5900 LampSite

- Editorial changes

  None

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

Trial Features

Trial features are features that are not yet ready for full commercial release for certain reasons. For example, the industry chain (terminals/CN) may not be sufficiently compatible. However, these features can still be used for testing purposes or commercial network trials. Anyone who desires to use the trial features shall contact Huawei and enter into a memorandum of understanding (MoU) with Huawei prior to an official application of such trial features. Trial features are not for sale in the current version but customers may try them for free.

Customers acknowledge and undertake that trial features may have a certain degree of risk due to absence of commercial testing. Before using them, customers shall fully understand not only the expected benefits of such trial features but also the possible impact they may exert on the network. In addition, customers acknowledge and undertake that since trial features are free, Huawei is not liable for any trial feature malfunctions or any losses incurred by using the trial features. Huawei does not promise that problems with trial features will be resolved in the current version. Huawei reserves the rights to convert trial features into commercial features in later R/C versions. If trial features are converted into commercial features in a later version, customers shall pay a licensing fee to obtain the relevant licenses prior to using the said commercial features. If a customer fails to purchase such a license, the trial feature(s) will be invalidated automatically when the product is upgraded.

2.2 Features in This Document

This document describes the following feature.

Feature ID	Feature Name	Chapter/Section
FOFD-081233	URLLC for Vehicle Communications (Trial)	**4 URLLC for Vehicle Communications**

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
URLLC for Vehicle Communications	Supported only in SA networking	**4 URLLC for Vehicle Communications**

Differences Between High Frequency Bands and Low Frequency Bands

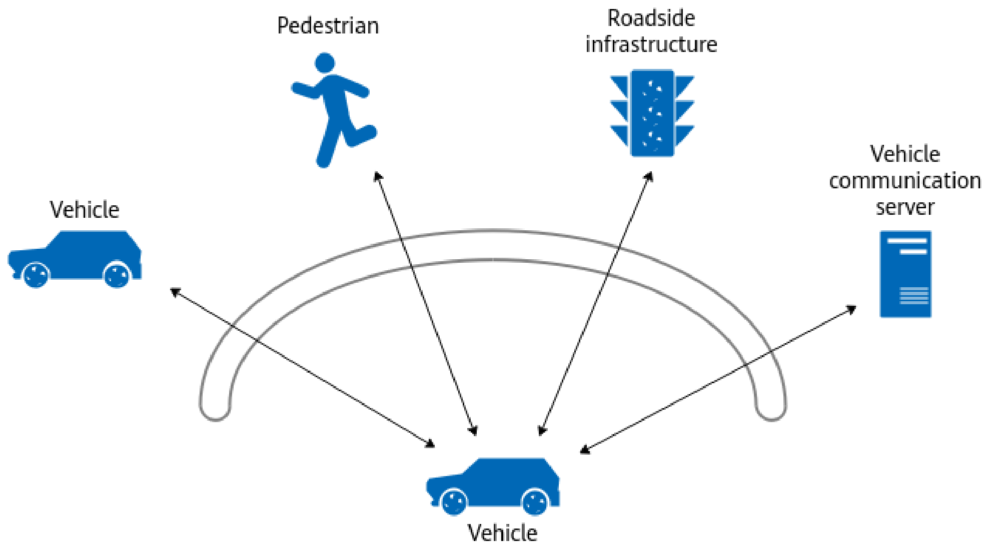
This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
URLLC for Vehicle Communications	Supported only in low frequency bands	**4 URLLC for Vehicle Communications**

3 Overview

Vehicle communication services include communications between vehicles, between vehicles and roadside infrastructure, between vehicles and networks, and between vehicles and pedestrians, as depicted in **Figure 3-1**. They provide real-time communications for road safety, traffic efficiency, and value-added applications such as intelligent transportation systems.

Figure 3-1 Vehicle communication services



Traditional vehicle communication services, carried on private networks via the PC5 interface, are developing at a slow pace. As such, operators expect 5G public networks to enable vehicle communication services via the Uu interface, thereby increasing the value of 5G public networks. However, existing public networks are not optimized for vehicle communication services, resulting in high latency and negatively impacting E2E service experience.

URLLC for Vehicle Communications provides latency assurance for 5QI-specific bearers to reduce the transmission delay of vehicle communication messages carried on public networks via the Uu interface and enable vehicle communication services such as vehicle-road synergy, vehicle-cloud synergy, and assisted driving. URLLC stands for ultra-reliable low-latency communication.

Table 3-1 and **Table 3-2** describe the message types and typical service models for vehicle communication services defined in *CSAE 157-2020 Cooperative Intelligent Transportation System — Vehicular Communication Application Layer Specification and Data Exchange Standard (Phase II)*. **Table 3-2** only provides reference values for typical scenarios. The actual values depend on the specific scenario. For example, the actual packet size for RSM is determined by the number of drive test objects.

Table 3-1 Message types for vehicle communication services

Message Type	Direction	Content
Road Side Message (RSM)	Downlink	Indicates roadside unit messages, including a list of target object information obtained after fusion sensing by a sensing engine on the data of radar cameras, that is, location and speed information of vehicles and pedestrians.
Signal Phase and Timing (SPaT)	Downlink	Indicates the phase and timing information of traffic signals.
Map	Downlink	Indicates map information, including lane lines, lane widths, and lane directions.
Road Side Information (RSI)	Downlink	Indicates roadside traffic information, including dynamic information such as traffic collisions, traffic jams, and speed violations, semi-persistent information such as road construction, and static information such as traffic signs (e.g. speed limit signs).
Radio Technical Commission for Maritime Services (RTCM)	Downlink	Indicates GNSS satellite information such as position, velocity, and time.
Basic Safety Message (BSM)	Uplink	Indicates basic information for traffic safety, including vehicle positions and speeds.

Table 3-2 Typical service models for vehicle communication services

Message Type	Interval (ms)	Packet Size (byte)
RSM	100	1800
SPaT	500	1000
Map	10000	2000
Dynamic RSI	100	500
Static RSI	1000	750

Message Type	Interval (ms)	Packet Size (byte)
Semi-persistent RSI	500	750
RTCM	1000	1020
BSM	100	360

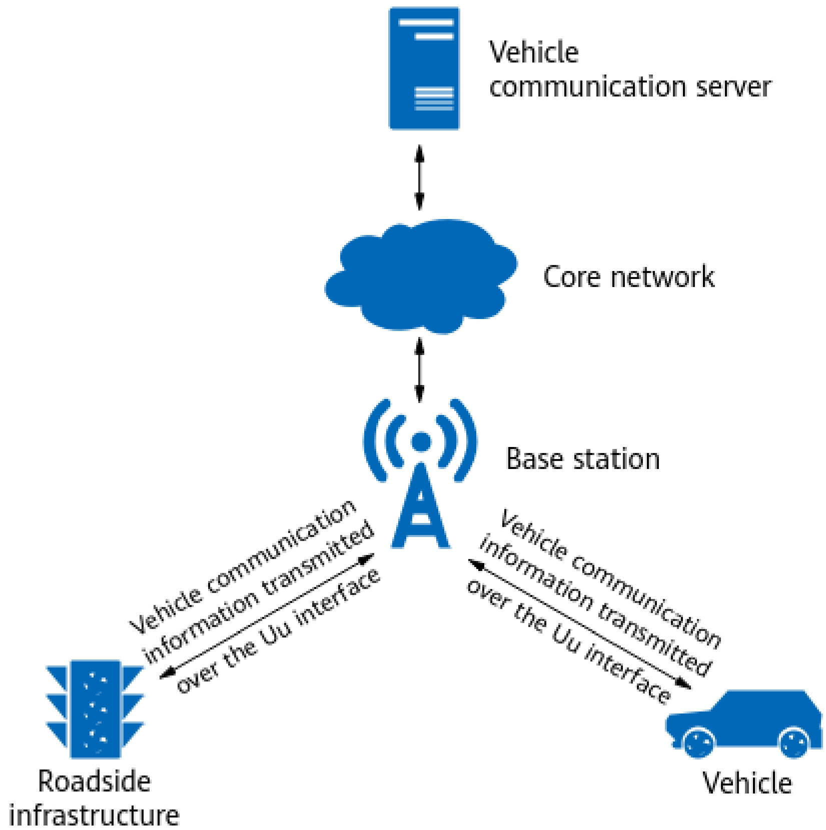
4 URLLC for Vehicle Communications

4.1 Principles

Basic Concepts

Figure 4-1 shows the typical architecture of URLLC for Vehicle Communications.

Figure 4-1 Typical architecture of URLLC for Vehicle Communications



URLLC for Vehicle Communications reduces the downlink delay of vehicle communication services by identifying vehicle communication UEs as URLLC UEs.

This function enables 3GPP-defined 5QI-specific bearers, such as 5QI 79 bearers, to carry vehicle communication services. For details about 3GPP-defined 5QIs, see section 5.7.4 "Standardized 5QI to QoS characteristics mapping" in 3GPP TS 23.501 V15.0.0.

QoS Management

QoS management is a mechanism that helps a network meet service quality requirements. The mechanism does this by implementing E2E coordination among all the involved network nodes, from the initiation to the response of a service. gNodeB QoS management includes radio QoS management (see *QoS Management*) and transmission QoS management (see *Transmission Resource Management*). To set up vehicle communication services, the base station and core network must configure 5QI-specific bearers with high priority for vehicle communication services. **Table 4-1** lists the 5QIs supporting vehicle communication services.

Table 4-1 5QIs supporting vehicle communication services

5QI	Default Priority Level	Packet Delay Budget	Packet Error Rate	Default Maximum Data Burst Volume	Default Averaging Window	Example Services
3	30	50 ms	10^{-3}	N/A	2000 ms	Real-time gaming and vehicle-to-everything (V2X) messages
79	65	50 ms	10^{-2}	N/A	N/A	V2X messages
83	22	10 ms	10^{-4}	1354 bytes	2000 ms	Discrete automation and V2X messages (vehicle platooning and advanced driving, such as collaborative lane changing with a low level of automation (LoA))
85	21	5 ms	10^{-5}	255 bytes	2000 ms	Electricity distribution (high voltage) and V2X messages (remote driving)
86	18	5 ms	10^{-4}	1354 bytes	2000 ms	V2X messages (advanced driving, such as vehicle platooning with a high LoA and collision avoidance)

UE Identification

In actual scenarios, a vehicle communication UE performs both vehicle communication and enhanced Mobile Broadband (eMBB) services, thus requiring multiple bearers with different 5QIs. With hybrid service scheduling enabled, the base station can identify vehicle communication UEs as URLLC UEs, so that URLLC can take effect for these UEs. Hybrid service scheduling is controlled by the **HYBRID_SERVICE_SCH_SW** option of the **NRDUCellAlgoSwitch**.***HighReliabilitySwitch*** parameter.

Latency Assurance

After a vehicle communication UE is identified as a URLLC UE, the low latency and high reliability function and service-differentiated low latency can be used to meet the latency requirements. For details about the principles of these functions, see *URLLC* and *Service-differentiated Experience Guarantee*. The downlink delay can be minimized only when all of the following functions are used together. In TDD, the communication requirements of assisted driving services can be met only when all of the following functions are used together.

- Low latency and high reliability

  The low latency and high reliability function improves service reliability by implementing data channel reliability and control channel reliability. It is controlled by the **HIGH_RELIABILITY_BASIC_SW** option of the **NRDUCellAlgoSwitch**.***HighReliabilitySwitch*** parameter. When this option is selected, data channel reliability and control channel reliability configurations can be performed to reduce the block error rate (BLER), thereby improving the reliability.

  Given that the low latency and high reliability function takes effect at the UE level, the following 5QI configurations must be performed so that this function can take effect for a UE carried on a 5QI-specific bearer:

  – Setting the **gNBDUMacParamGroup**.***SchedulingType*** parameter to **HIGH_RELIABILITY_SCH** to configure high-reliability-oriented scheduling for the 5QI-specific bearer

  – Setting the **gNBQciBearer**.***ServiceType*** parameter to **URLLC_HIGH_RELIABILITY_SERVICE** to enable the 5QI-specific bearer to carry high-reliability services

  During UE inter-frequency measurements, the performance of vehicle communication services deteriorates due to the start of gap-assisted measurements. To minimize the number of inter-frequency measurements to be triggered, you are advised to configure the **NRCellInterFHoMeaGrp**.***CovInterFreqA2RsrpThld***, **NRCellInterFHoMeaGrp**.***FreqPriInterFA1RsrpThld***, and **NRCellInterFHoMeaGrp**.***FreqPriInterFA2RsrpThld*** parameters associated with the 5QI-specific bearers for vehicle communication services based on the actual network plan.

  During discontinuous reception (DRX), UEs cannot receive downlink services in the sleep time. This results in waiting time for scheduling, affecting the performance of vehicle communication services. To avoid this, you are advised not to enable DRX for vehicle communication service bearers. That is, you should set the **NRCellQciBearer**.***DrxParamGroupId*** or

NRDUCellQciBearer.***DuDrxParamGroupId*** parameter associated with the 5QI-specific bearers for vehicle communication services to **255**.

Specifically, the following functions are involved.

- Resource reservation

  This function is controlled by the **RB_DYNAMIC_CONTROL_SW** option of the **NRDUCellAlgoSwitch**.***HighReliabilitySwitch*** parameter. For vehicle communication services, the downlink delay requirements can be met by setting the **NRDUCellRes**.***ResStrategyType***, **NRDUCellRes**.***DlRbMinRatio***, **NRDUCellRes**.***DlRbMaxRatio***, and **NRDUCellRes**.***DlMinRbStartPos*** parameters.

- OLLA algorithm optimization

  For vehicle communication services, the downlink delay requirements can be met by setting the **gNBDUSchParamGroup**.***DlInitialMcsAdjValue***, **gNBDUSchParamGroup**.***DlAmcAdjustStep***, **gNBDUSchParamGroup**.***DlMaxMcs***, and **gNBDUSchParamGroup**.***DlMinMcs*** parameters.

- HARQ retransmission optimization

  For vehicle communication services, the delay requirements can be met by setting the **gNBDUSchParamGroup**.***DlMaxHarqRetransNum*** and **gNBDUSchParamGroup**.***UlMaxHarqRetransNum*** parameters.

- Target PDCCH IBLER optimization

  For vehicle communication services, the downlink delay requirements can be met by setting the **gNBDUSchParamGroup**.***PdcchBlerTarget*** parameter.

- PDCCH CCE optimization

  For vehicle communication services, the downlink delay requirements can be met by setting the **NRDUCellPdcch**.***OccupiedSymbolNum*** parameter and the **UL_DL_CCE_RATIO_ADAPT_SW** option of the **NRDUCellPdcch**.***PdcchAlgoSwitch*** parameter.

- PDCCH aggregation level adjustment mechanism optimization

  For vehicle communication services, the downlink delay requirements can be met by setting the **gNBDUSchParamGroup**.***PdcchMinAggLvl*** parameter.

- PDCP out-of-order delivery

  For vehicle communication services, the processing time at receivers can be reduced through PDCP out-of-order delivery. If a receiver supports PDCP out-of-order delivery, any received out-of-order packets are not reordered at the PDCP layer. Instead, they are directly delivered to the upper layer for processing, which avoids the delay caused by PDCP reordering. This function includes PDCP out-of-order delivery on both the base station and UE sides, which are configured as follows:

  - If a bearer is being set up for a UE and the **gNBPdcpParamGroup**.***GnbPdcpOutOfOrderDeliveryInd*** parameter is set to **TRUE_WITHOUT_UE_CAPB** or **TRUE_WITH_UE_CAPB**, the PDCP entity of the bearer on the base station side directly delivers received packets to the core network without reordering them. In this case, you can set the **gNBPdcpParamGroup**.***gNBPdcpReorderingTimer*** parameter to specify the length of the gNodeB's PDCP reordering timer. The timer starts when the base station receives out-of-order packets. Once the timer expires, the base station forwards the received PDCP packets to the upper layer and discards any missing packets that have not been received.

- If a UE supports PDCP out-of-order delivery and the **gNBPdcpParamGroup**.***UePdcpOutOfOrderDeliveryInd*** parameter is set to **TRUE**, the PDCP entity of the UE is configured to directly deliver received packets to the application layer without reordering them.

– Rank configuration for low latency and high reliability

UE-reported ranks may be inaccurate due to the complex network environment. When the base station performs scheduling based on a UE-reported rank that is higher than expected, bit errors occur and the service delay increases. Rank configuration for low latency and high reliability can specify the maximum rank used in the uplink and downlink for vehicle communication UEs. This function is controlled by the **gNBDUSchParamGroup**.***UrllcMaxRank*** parameter.

- If this parameter is set to **NOT_CONFIG**, this parameter does not take effect. In this case, the maximum rank to be used in the uplink and downlink for vehicle communication UEs is not configured.
- If this parameter is set to **RANK1**, rank 1 is the maximum rank to be used in the uplink and downlink for vehicle communication UEs.
- If this parameter is set to **RANK2**, rank 2 is the maximum rank to be used in the uplink and downlink for vehicle communication UEs.

The value **RANK2** is recommended when frequency-domain resources are insufficient, and the value **RANK1** is recommended in other scenarios.

- Service-differentiated low latency

URLLC for Vehicle Communications uses service-differentiated low latency and its sub-functions to implement 5QI-specific service assurance. This results in faster scheduling, reduced BLER, and fewer retransmissions, ultimately leading to shorter air interface delay. Service-differentiated low latency is controlled by the **SERV_DIFF_LOW_LATENCY_SW** option of the **NRDUCellAlgoSwitch**.***LowLatencySwitch*** parameter.

– QCI-based uplink preallocation

The gNodeB allows for the differentiation of preallocation parameter group configuration between QCI-specific bearers. For vehicle communication services, the delay requirements can be met by selecting the **UL_PREALLOCATION_QCI_SW** option of the **gNBDUSchParamGroup**.***ScheduleOptSwitch*** parameter and setting the **gNBDUSchParamGroup**.***UlPreallocationInterval***, **gNBDUSchParamGroup**.***UlPreallocationDataVolume***, and **NRDUCellPusch**.***PreschUeNumUpperLimit*** parameters to appropriate values.

– QCI-specific SR period configuration

A scheduling request (SR) period can be customized for services carried on a QCI-specific bearer. For vehicle communication services, the delay requirements can be met by selecting the **SR_PERIOD_QCI_SW** option of the **gNBDUSchParamGroup**.***ScheduleOptSwitch*** parameter and setting the **gNBDUSchParamGroup**.***SrPeriod*** parameter to an appropriate value.

– QCI-specific first-packet size configuration for SR-based scheduling

  A first-packet size can be customized for a QCI-specific bearer in SR-based scheduling. For vehicle communication services, the delay requirements can be met by setting the **gNBDUSchParamGroup**.***SrSchFirstTbSize*** parameter to an appropriate value.

– QCI-specific scheduling based on the target IBLER

  Uplink and downlink target initial BLERs (IBLERs) can be configured for a specific QCI. For vehicle communication services, the delay requirements can be met by selecting the **UL_IBLER_QCI_SW** and **DL_IBLER_QCI_SW** options of the **gNBDUSchParamGroup**.***ScheduleOptSwitch*** parameter and setting the **gNBDUSchParamGroup**.***UlTargetIbler*** and **gNBDUSchParamGroup**.***DlTargetIbler*** parameters to appropriate values.

– New policy for service differentiation

  For a UE with multiple QCI-specific bearers, the implementation of a QCI-based uplink preallocation function (either basic or smart) depends on different policies described below.

  - Policy 1 applies when the **SERV_DIFF_POLICY_SW** option of the **NRDUCellAlgoSwitch**.***LowLatencySwitch*** parameter is deselected. With this policy, a QCI-based uplink preallocation function (either basic or smart) can work on the UE only if the function is enabled for the UE's QCI of the highest priority. This means that if the function is enabled for other QCIs of the UE, the function does not work on the UE.

  - Policy 2 applies when the **SERV_DIFF_POLICY_SW** option of the **NRDUCellAlgoSwitch**.***LowLatencySwitch*** parameter is selected. With this policy, a QCI-based uplink preallocation function (either basic or smart) can work on the UE in accordance with the following configurations as long as the function is enabled for any QCI of the UE.
    - Preallocation interval: Among the preallocation intervals configured for all the QCIs for which QCI-based uplink basic preallocation or QCI-based uplink smart preallocation is enabled, the minimum preallocation interval applies.
    - Uplink preallocation data volume: Among the uplink preallocation data volumes configured for all the QCIs for which QCI-based uplink basic preallocation or QCI-based uplink smart preallocation is enabled, the maximum uplink preallocation data volume applies.
    - Smart preallocation duration: Among the smart preallocation durations configured for all the QCIs for which QCI-based uplink smart preallocation is enabled, the maximum smart preallocation duration applies.

  For vehicle communication services, the delay requirements can be met by selecting the **SERV_DIFF_POLICY_SW** option of the **NRDUCellAlgoSwitch**.***LowLatencySwitch*** parameter.

4.2 Network Analysis

4.2.1 Benefits

URLLC for Vehicle Communications can reduce the communication delay of vehicle communication UEs and improve data transmission reliability. You are advised to enable this function when all of the following conditions are met. In TDD, this function can meet the communication requirements of assisted driving services when all of the following conditions are met.

- Scenarios: On outdoor roads, the vehicle speed is below 80 km/h, in-vehicle terminals have a synchronization signal and PBCH block (SSB) reference signal received power (RSRP) of at least –93 dBm and an SSB signal to interference plus noise ratio (SINR) of at least –3 dB, and the downlink physical resource block (PRB) usage is less than 20%.
- Specifications: In each cell, the number of vehicle communication UEs is no more than 10.
- Core network configuration: Vehicle communication services are assigned high-priority 5QI-specific bearers, such as 5QI 79 bearers.
- Latency between the base station and upper-layer NEs (typically the vehicle communication server): must be less than 5 ms.
- Terminal processing latency: must be less than 2 ms.

4.2.2 Impacts

The impacts between this function and other functions are described in **4.3.2.3 Function Impacts**.

After URLLC for Vehicle Communications is enabled, the following indicators will be affected.

- The PDCCH CCE usage (**N.CCE.Used.Avg**/**N.CCE.Avail.Avg**) may increase.
- The total number of scheduled UEs in the downlink in a cell (**N.User.Schedule.Dl.Sum**) may decrease.
- The average downlink cell throughput (**Cell Downlink Average Throughput (DU)**) may decrease.
- The average downlink UE throughput (**User Downlink Average Throughput (DU)**) may change.
- The downlink PRB usage (**Downlink Resource Block Utilizing Rate (DU)**) may increase.
- The average downlink MCS index may decrease. (Average downlink MCS index = SUM(*k* x **N.ChMeas.PDSCH.MCS.*k***)/SUM(**N.ChMeas.PDSCH.MCS.*k***), where *k* ranges from 0 to 28)
- The downlink IBLER may decrease. (Downlink IBLER = (**N.DL.SCH.QPSK.ErrTB.Ibler** + **N.DL.SCH.16QAM.ErrTB.Ibler** + **N.DL.SCH.64QAM.ErrTB.Ibler** + **N.DL.SCH.256QAM.ErrTB.Ibler**)/(**N.DL.SCH.QPSK.TB** + **N.DL.SCH.16QAM.TB** + **N.DL.SCH.64QAM.TB** + **N.DL.SCH.256QAM.TB**) x 100%)

- The downlink residual BLER (RBLER) may decrease. (Downlink RBLER = (**N.DL.SCH.QPSK.ErrTB.Rbler** + **N.DL.SCH.16QAM.ErrTB.Rbler** + **N.DL.SCH.64QAM.ErrTB.Rbler** + **N.DL.SCH.256QAM.ErrTB.Rbler**)/ (**N.DL.SCH.QPSK.TB** + **N.DL.SCH.16QAM.TB** + **N.DL.SCH.64QAM.TB** + **N.DL.SCH.256QAM.TB**) x 100%)
- The average number of used downlink PRBs (**N.PRB.DL.Used.Avg**) fluctuates.
- The average number of PDCCH CCEs used for downlink DCI (**N.CCE.DLDCI.Used.Avg**) may decrease.

4.3 Requirements

4.3.1 Licenses

There are no license requirements for trial functions.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Service-differentiated low latency	**SERV_DIFF_LOW_LATENCY_SW** option of the **NRDUCellAlgoSwitch**.***LowLatencySwitch*** parameter	*Service-differentiated Experience Guarantee*	URLLC for Vehicle Communications takes effect only after service-differentiated low latency is enabled.
Low-frequency TDD	Low latency and high reliability	**HIGH_RELIABILITY_BASIC_SW** option of the **NRDUCellAlgoSwitch**.***HighReliabilitySwitch*** parameter	*URLLC*	URLLC for Vehicle Communications takes effect only after low latency and high reliability is enabled.

4.3.2.2 Mutually Exclusive Functions

As described in **4.1 Principles**, URLLC for Vehicle Communications uses the low latency and high reliability function and service-differentiated low latency to meet the latency requirements. For details about the mutually exclusive functions of these functions, see *URLLC* and *Service-differentiated Experience Guarantee*.

4.3.2.3 Function Impacts

As described in **4.1 Principles**, URLLC for Vehicle Communications uses the low latency and high reliability function and service-differentiated low latency to meet the latency requirements. For details about the function impacts of these functions, see *URLLC* and *Service-differentiated Experience Guarantee*.

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

Baseband processing units: Only UBBPg and UBBPh series boards support this function.

Main control boards: Only UMPTe, UMPTg, UMPTha, and UMPThb series boards support this function.

RF Modules

In NR TDD, only NR TDD-capable 4T4R/8T8R/32T32R/64T64R RF modules that work in low frequency bands support this function.

4.3.4 Cells

In NR TDD, the cell bandwidth must be 60 MHz, 80 MHz, or 100 MHz.

4.3.5 Others

Core network: The latency between the base station and upper-layer NEs (typically the vehicle communication server) must be less than 5 ms. Vehicle communication services must be assigned high-priority 5QI-specific bearers, such as 5QI 79 bearers.

UE: For PDCP out-of-order delivery, UEs must have the capability to support it. The **outOfOrderDelivery** field in the *PDCP-Parameters* IE of RRC messages indicates whether a UE has this capability. The value of this field can be queried in the RRC_UE_CAP_INFO message over the air interface in the access procedure.

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

Table 4-2 describes the parameters used for function activation. No parameters are involved in function optimization.

Table 4-2 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
High Reliability Switch	**NRDUCellAlgoSwitch**.***HighReliabilitySwitch***	HIGH_RELIABILITY_BASIC_SW	Select this option for vehicle communications.
Service Type	**gNBQciBearer**.***ServiceType***	None	Use the value **URLLC_HIGH_RELIABILITY_SERVICE** for vehicle communications.
Scheduling Type	**gNBDUMacParamGroup**.***SchedulingType***	None	Use the value **HIGH_RELIABILITY_SCH** for vehicle communications.
High Reliability Switch	**NRDUCellAlgoSwitch**.***HighReliabilitySwitch***	RB_DYNAMIC_CONTROL_SW	Select this option for vehicle communications.
High Reliability Switch	**NRDUCellAlgoSwitch**.***HighReliabilitySwitch***	HYBRID_SERVICE_SCH_SW	Select this option for vehicle communications.
Cov Inter-freq Handover A2 RSRP Threshold	**NRCellInterFHoMeaGrp**.***CovInterFreqA2RsrpThld***	None	Use the value **-110** for vehicle communications.
Freq Priority Inter-freq A1 RSRP Threshold	**NRCellInterFHoMeaGrp**.***FreqPriInterFA1RsrpThld***	None	Use the value **-31** for vehicle communications.
Freq Priority Inter-freq A2 RSRP Threshold	**NRCellInterFHoMeaGrp**.***FreqPriInterFA2RsrpThld***	None	Use the value **-103** for vehicle communications.

Parameter Name	Parameter ID	Option	Setting Notes
DRX Parameter Group ID	**NRCellQciBearer**.***DrxParamGroupId***	None	Use the value **255** for vehicle communications.
DU DRX Parameter Group ID	**NRDUCellQciBearer**.***DuDrxParamGroupId***	None	Use the value **255** for vehicle communications.
Resource Usage Object	**NRDUCellRes**.***ResUsageObject***	None	Use the value **HIGH_RELIABILITY** for vehicle communications.
Resource Strategy Type	**NRDUCellRes**.***ResStrategyType***	None	Use the value **TTI_ONLY** for vehicle communications.
Downlink RB Min Ratio	**NRDUCellRes**.***DlRbMinRatio***	None	Use the value **30** for vehicle communications.
Downlink RB Max Ratio	**NRDUCellRes**.***DlRbMaxRatio***	None	Use the value **50** for vehicle communications.
Downlink Min RB Start Position	**NRDUCellRes**.***DlMinRbStartPos***	None	Use the value **0** for vehicle communications.
Downlink Initial MCS Adjust Value	**gNBDUSchParamGroup**.***DlInitialMcsAdjValue***	None	Use the recommended value.
Downlink AMC Adjustment Step	**gNBDUSchParamGroup**.***DlAmcAdjustStep***	None	Use the recommended value.
Downlink Maximum MCS	**gNBDUSchParamGroup**.***DlMaxMcs***	None	Use the recommended value.
Downlink Minimum MCS	**gNBDUSchParamGroup**.***DlMinMcs***	None	Use the recommended value.
Downlink Maximum HARQ Retrans Number	**gNBDUSchParamGroup**.***DlMaxHarqRetransNum***	None	Use the value **2** for vehicle communications.
Uplink Maximum HARQ Retrans Number	**gNBDUSchParamGroup**.***UlMaxHarqRetransNum***	None	Use the default value in TDD for vehicle communications.

Parameter Name	Parameter ID	Option	Setting Notes
PDCCH BLER Target	**gNBDUSchParamGroup**.***PdcchBlerTarget***	None	Use the recommended value.
Occupied Symbol Number	**NRDUCellPdcch**.***OccupiedSymbolNum***	None	For vehicle communications, use the value **1SYM** when the PDCCH is not congested; use the value **2SYM** when the PDCCH is congested.
PDCCH Algorithm Switch	**NRDUCellPdcch**.***PdcchAlgoSwitch***	UL_DL_CCE_RATIO_ADAPT_SW	Select this option for vehicle communications.
PDCCH Min Aggregation Level	**gNBDUSchParamGroup**.***PdcchMinAggLvl***	None	Use the value **AGGLVL2** for vehicle communications.
Low Latency Switch	**NRDUCellAlgoSwitch**.***LowLatencySwitch***	SERV_DIFF_LOW_LATENCY_SW	Select this option for vehicle communications.
Schedule Option Switch	**gNBDUSchParamGroup**.***ScheduleOptSwitch***	UL_PREALLOCATION_QCI_SW	Select this option for vehicle communications.
Uplink Preallocation Interval	**gNBDUSchParamGroup**.***UlPreallocationInterval***	None	Use the value **10** in TDD for vehicle communications.
Uplink Preallocation Data Volume	**gNBDUSchParamGroup**.***UlPreallocationDataVolume***	None	Use the value **1500** for vehicle communications.
Prescheduled UE Num Upper Limit	**NRDUCellPusch**.***PreschUeNumUpperLimit***	None	Use the value **16** in TDD for vehicle communications.
Schedule Option Switch	**gNBDUSchParamGroup**.***ScheduleOptSwitch***	SR_PERIOD_QCI_SW	Select this option for vehicle communications.
SR Period	**gNBDUSchParamGroup**.***SrPeriod***	None	Use the value **20SLOT** for vehicle communications.

Parameter Name	Parameter ID	Option	Setting Notes
SR-based Scheduling First TB Size	**gNBDUSchParamGroup**.***SrSchFirstTbSize***	None	For vehicle communications, you are advised to set this parameter based on the actual service packet size.
Schedule Option Switch	**gNBDUSchParamGroup**.***ScheduleOptSwitch***	UL_IBLER_QCI_SW	Select this option for vehicle communications.
Uplink Target IBLER	**gNBDUSchParamGroup**.***UlTargetIbler***	None	Use the value **10000** in TDD for vehicle communications.
Schedule Option Switch	**gNBDUSchParamGroup**.***ScheduleOptSwitch***	DL_IBLER_QCI_SW	Select this option for vehicle communications.
Downlink Target IBLER	**gNBDUSchParamGroup**.***DlTargetIbler***	None	Use the value **1000** for vehicle communications.
Low Latency Switch	**NRDUCellAlgoSwitch**.***LowLatencySwitch***	SERV_DIFF_POLICY_SW	Select this option for vehicle communications.
gNodeB PDCP Out Of Order Delivery Ind	**gNBPdcpParamGroup**.***GnbPdcpOutOfOrderDeliveryInd***	None	Use the value **TRUE_WITHOUT_UE_CAPB** for vehicle communications.
gNodeB PDCP Reordering Timer	**gNBPdcpParamGroup**.***gNBPdcpReorderingTimer***	None	Use the value **MS10** for vehicle communications.
UE PDCP Out Of Order Delivery Ind	**gNBPdcpParamGroup**.***UePdcpOutOfOrderDeliveryInd***	None	Use the value **TRUE** for vehicle communications.
URLLC Max Rank	**gNBDUSchParamGroup**.***UrllcMaxRank***	None	For vehicle communications, the value **RANK2** is recommended when frequency-domain resources are insufficient, and the value **RANK1** is recommended in other scenarios.

4.4.1.2 Using MML Commands

NOTICE

The related functions take effect five to ten minutes after the relevant parameter configurations in the **gNBDUSchParamGroup** and **NRDUCellQciBearer** MOs are modified.

When resource reservation is used to guarantee data resource availability for high-reliability services and resources are allocated on a per resource block group (RBG) basis, the maximum number of RBs available for each UE must be greater than or equal to the number of RBs included in one RBG, regardless of whether the UE performs high-reliability services or not. That is, both of the following conditions must be met:

- Number of RBs corresponding to **NRDUCellRes**.***DlRbMaxRatio*** ≥ Number of RBs in an RBG
- Total number of available PDSCH RBs in a cell – Number of RBs corresponding to **NRDUCellRes**.***DlRbMinRatio*** for UEs performing high-reliability services ≥ Number of RBs in an RBG

Before enabling the low latency and high reliability function, ensure that the bearers corresponding to the QCIs used for high-reliability services in all cells of a base station are configured as high-reliability bearers. Otherwise, the low latency and high reliability function does not take effect for the QCI-specific bearers.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to **4.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

Basic configuration

```
//Turning on the switch for basic high-reliability functions
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, HighReliabilitySwitch=HIGH_RELIABILITY_BASIC_SW-1;
//Turning on the switch for hybrid service scheduling
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, HighReliabilitySwitch=HYBRID_SERVICE_SCH_SW-1;
//(Recommended) Using 5QI 79 bearers for vehicle communication services (5QI 79 bearers are set up by
default and do not require configuration on the base station side. However, 5QI 79 bearer setup requires
support from the core network.) For all cells of a base station, if the bearers corresponding to a 5QI need to
be configured as high-reliability bearers, the gNBQciBearer.ServiceType parameter must be set to
URLLC_HIGH_RELIABILITY_SERVICE. Otherwise, the low latency and high reliability function does not take
effect for the 5QI-specific bearers.
MOD GNBQCIBEARER: Qci=79, ServiceType=URLLC_HIGH_RELIABILITY_SERVICE;
//Configuring the bearers as high-reliability bearers and associating them with a 5QI
MOD GNBDUMACPARAMGROUP: MacParamGroupId=79, SchedulingType=HIGH_RELIABILITY_SCH;
MOD NRDUCELLQCIBEARER: NrDuCellId=0, Qci=79, MacParamGroupId=79;
//Configuring the RSRP threshold for event A2 related to coverage-based inter-frequency handover and the
RSRP thresholds for events A1 and A2 related to frequency-priority-based inter-frequency handover
MOD NRCELLQCIBEARER: NrCellId=0, Qci=79, InterFreqHoMeasGroupId=0;
```

```
MOD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGroupId=0, CovInterFreqA2RsrpThld=-110, FreqPriInterFA1RsrpThld=-31, FreqPriInterFA2RsrpThld=-103;
//Disabling DRX
MOD NRCELLQCIBEARER: NrCellId=0, Qci=79, DrxParamGroupId=255;
MOD NRDUCELLQCIBEARER: NrDuCellId=0, Qci=79, DuDrxParamGroupId=255;
```

Resource reservation

```
//Enabling resource reservation for high-reliability services
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, HighReliabilitySwitch=RB_DYNAMIC_CONTROL_SW-1;
ADD NRDUCELLRES: NrDuCellId=0, CellResId=0, ResUsageObject=HIGH_RELIABILITY, ResStrategyType=TTI_ONLY, DlMinRbStartPos=0, DlRbMinRatio=30, DlRbMaxRatio=50;
MOD NRDUCELLQCIBEARER: NrDuCellId=0, Qci=79, CellResId=0;
```

OLLA algorithm optimization

```
//Adding a gNodeB DU scheduling parameter group
ADD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1;
//Enabling OLLA algorithm optimization
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1, DlInitialMcsAdjValue=-4, DlAmcAdjustStep=10, DlMaxMcs=28, DlMinMcs=0;
//Mapping the gNodeB DU scheduling parameter group to the 5QI-specific bearers
MOD NRDUCELLQCIBEARER: NrDuCellId=0, Qci=79, ScheduleParamGroupId=1;
```

HARQ retransmission optimization

```
//Enabling HARQ retransmission optimization
//Adding a gNodeB DU scheduling parameter group
ADD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1;
//Configuring the maximum number of HARQ retransmissions (TDD)
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1, DlMaxHarqRetransNum=2, UlMaxHarqRetransNum=2;
//Mapping the gNodeB DU scheduling parameter group to the 5QI-specific bearers
MOD NRDUCELLQCIBEARER: NrDuCellId=0, Qci=79, ScheduleParamGroupId=1;
```

Target PDCCH IBLER optimization

```
//Adding a gNodeB DU scheduling parameter group
ADD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1;
//Configuring the target PDCCH IBLER
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1, PdcchBlerTarget=2;
//Mapping the gNodeB DU scheduling parameter group to the 5QI-specific bearers
MOD NRDUCELLQCIBEARER: NrDuCellId=0, Qci=79, ScheduleParamGroupId=1;
```

PDCCH CCE optimization

```
//Setting the parameter specifying the number of symbols occupied by the PDCCH to 1SYM when the PDCCH is not congested or 2SYM when the PDCCH is congested (Modification of this parameter will cause the cell to automatically restart.)
MOD NRDUCELLPDCCH: NrDuCellId=0, OccupiedSymbolNum=1SYM;
//Turning on the switch for PDCCH uplink-to-downlink CCE ratio adaptation
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoSwitch=UL_DL_CCE_RATIO_ADAPT_SW-1;
```

PDCCH aggregation level adjustment mechanism optimization

```
//Adding a gNodeB DU scheduling parameter group
ADD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1;
//Configuring the minimum PDCCH aggregation level
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1,PdcchMinAggLvl=AGGLVL2;
//Mapping the gNodeB DU scheduling parameter group to the 5QI-specific bearers
MOD NRDUCELLQCIBEARER: NrDuCellId=0, Qci=79, ScheduleParamGroupId=1;
```

Service-differentiated low latency

```
//Turning on the switch for service-differentiated low latency
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, LowLatencySwitch=SERV_DIFF_LOW_LATENCY_SW-1;
//Adding a gNodeB DU scheduling parameter group
ADD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1;
//(TDD) Enabling QCI-based uplink preallocation
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1, ScheduleOptSwitch=UL_PREALLOCATION_QCI_SW-1, UlPreallocationInterval=10, UlPreallocationDataVolume=1500;
```

```
MOD NRDUCELLPUSCH: NrDuCellId=0, PreschUeNumUpperLimit=16;
//Enabling QCI-specific SR period configuration
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1, ScheduleOptSwitch=SR_PERIOD_QCI_SW-1,
SrPeriod=20SLOT;
//Enabling QCI-specific first-packet size configuration for SR-based scheduling
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1, SrSchFirstTbSize=1000;
//(TDD) Enabling QCI-specific scheduling based on the target IBLER
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1,
ScheduleOptSwitch=UL_IBLER_QCI_SW-1&DL_IBLER_QCI_SW-1, DlTargetIbler=1000, UlTargetIbler=10000;
//Enabling a new policy for service differentiation
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, LowLatencySwitch=SERV_DIFF_POLICY_SW-1;
//Mapping the gNodeB DU scheduling parameter group to the 5QI-specific bearers
MOD NRDUCELLQCIBEARER: NrDuCellId=0, Qci=79, ScheduleParamGroupId=1;
```

PDCP out-of-order delivery

```
//Enabling PDCP out-of-order delivery in acknowledged mode (AM) or unacknowledged mode (UM)
//Taking AM as an example
MOD NRCELLQCIBEARER: NrCellId=0, Qci=79, AmPdcpParamGroupId=13;
MOD GNBPDCPPARAMGROUP: PdcpParamGroupId=13,
GnbPdcpOutOfOrderDeliveryInd=TRUE_WITHOUT_UE_CAPB, gNBPdcpReorderingTimer=MS10,
UePdcpOutOfOrderDeliveryInd=TRUE;
//Taking UM as an example
MOD NRCELLQCIBEARER: NrCellId=0, Qci=79, UmPdcpParamGroupId=2;
MOD GNBPDCPPARAMGROUP: PdcpParamGroupId=2,
GnbPdcpOutOfOrderDeliveryInd=TRUE_WITHOUT_UE_CAPB, gNBPdcpReorderingTimer=MS10,
UePdcpOutOfOrderDeliveryInd=TRUE;
```

Rank configuration for low latency and high reliability

```
//Setting the maximum rank for URLLC (RANK2 is recommended when frequency-domain resources are
insufficient and RANK1 is recommended in other scenarios)
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1, UrllcMaxRank=RANK1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

Resource reservation

```
//Disabling resource reservation for high-reliability services
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, HighReliabilitySwitch=RB_DYNAMIC_CONTROL_SW-0;
```

PDCCH CCE optimization

```
//Turning off the switch for PDCCH uplink-to-downlink CCE ratio adaptation
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoSwitch=UL_DL_CCE_RATIO_ADAPT_SW-0;
```

PDCCH aggregation level adjustment mechanism optimization

```
//Setting the parameter specifying the minimum PDCCH aggregation level to NOT_CONFIG
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1,PdcchMinAggLvl=NOT_CONFIG;
```

Service-differentiated low latency

```
//Disabling QCI-based uplink preallocation
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1,
ScheduleOptSwitch=UL_PREALLOCATION_QCI_SW-0;
//Disabling QCI-specific SR period configuration
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1, ScheduleOptSwitch=SR_PERIOD_QCI_SW-0;
//Disabling QCI-specific first-packet size configuration for SR-based scheduling
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1, SrSchFirstTbSize=0;
//Disabling QCI-specific scheduling based on the target IBLER
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1,
ScheduleOptSwitch=UL_IBLER_QCI_SW-0&DL_IBLER_QCI_SW-0;
//Disabling the new policy for service differentiation
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, LowLatencySwitch=SERV_DIFF_POLICY_SW-0;
```



```
//Turning off the switch for service-differentiated low latency
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, LowLatencySwitch=SERV_DIFF_LOW_LATENCY_SW-0;
```

PDCP out-of-order delivery

```
//Disabling PDCP out-of-order delivery
//Taking AM as an example
MOD GNBPDCPPARAMGROUP: PdcpParamGroupId=13, GnbPdcpOutOfOrderDeliveryInd=FALSE,
UePdcpOutOfOrderDeliveryInd=FALSE;
//Taking UM as an example
MOD GNBPDCPPARAMGROUP: PdcpParamGroupId=2, GnbPdcpOutOfOrderDeliveryInd=FALSE,
UePdcpOutOfOrderDeliveryInd=FALSE;
```

Rank configuration for low latency and high reliability

```
//Disabling rank configuration for low latency and high reliability
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1, UrllcMaxRank=NOT_CONFIG;
```

Basic configuration

```
//Turning off the switch for hybrid service scheduling
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, HighReliabilitySwitch=HYBRID_SERVICE_SCH_SW-0;
//Turning off the switch for basic high-reliability functions
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, HighReliabilitySwitch=HIGH_RELIABILITY_BASIC_SW-0;
```

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

Perform the following steps to verify whether URLLC for Vehicle Communications has taken effect (taking 5QI 79 as an example):

Step 1 Observe the average downlink RLC packet loss rate for UEs carried on 5QI 79 bearers.

Step 2 After the average downlink RLC packet loss rate for UEs carried on 5QI 79 bearers remains stable for a period of time, enable URLLC for Vehicle Communications.

Step 3 If the average downlink RLC packet loss rate for UEs carried on 5QI 79 bearers decreases after the function has been enabled for a period of time, this function has taken effect.

----End

Average downlink RLC packet loss rate for UEs carried on 5QI-specific bearers = **N.RLC.DL.TrfSDU.PktUuTrans.Loss.DuCell5QI**/ **N.RLC.DL.TrfSDU.RxPackets.DuCell5QI**

4.4.3 Network Monitoring

The indicators listed in **4.2 Network Analysis** can be used to monitor the benefits and impacts of this function.

5 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

Step 1 Open the EXCEL file of parameter reference.

Step 2 On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

Step 1 Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

6 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

7 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

8 Reference Documents

- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- 3GPP TS 23.501: "System Architecture for the 5G System"
- *CSAE 157-2020 Cooperative Intelligent Transportation System — Vehicular Communication Application Layer Specification and Data Exchange Standard (Phase II)*
- *URLLC*
- *QoS Management*
- *Transmission Resource Management*
- *Service-differentiated Experience Guarantee*

